

Linear_regression

April 29, 2025

0.1 Linear Regression

In this project, I focus on applying **Linear Regression** to predict rental prices for residential properties in Kuala Lumpur and Selangor, Malaysia.

The dataset was sourced from [Kaggle](#), containing around 19000 rental listings scraped from mudah.my, one of Malaysia's major property platforms

0.2 Dataset Summary

The dataset includes 13 features and 1 target variable:

Target: `monthly_rent` – monthly rental price in Malaysian Ringgit (RM)

Key Features:

`ads_id`: the listing ids (unique)

`prop_name`: name of the building/ property

`completion_year`: completion/ established year of the property

`location`: property location in Kuala Lumpur region

`property_type`: property type such as apartment, condominium, flat, duplex, studio, etc

`rooms`: number of rooms in the unit

`parking`: number of parking space for the unit

`bathroom`: number of bathrooms in the unit

`size`: total area of the unit in square feet

`furnished`: furnishing status of the unit (fully, partial, non-furnished)

`facilities`: main facilities available

`additional_facilities`: additional facilities (proximity to attraction area, mall, school, shopping, railways, etc)

##Methodology: Linear Regression

To build the linear regression model, I followed these steps:

Preprocessing:

Removed irrelevant columns (`ads_id`, `prop_name`, etc.)

Dropped missing values in critical features

Converted categorical variables (location, property_type, furnished) into dummy variables for regression

Train-Test Split:

Randomly split the dataset into training (80%) and test (20%) sets

Model Training:

Used the LinearRegression() model from scikit-learn to fit the training data

Measured training time to assess computational efficiency

Evaluation:

Predicted rent prices using the training set

Evaluated model performance using R^2 Score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE)

Visualization:

Created a scatter plot of actual vs predicted rent values on the training set

Added a reference line for perfect prediction ($y = x$)

```
[1]: import pandas as pd
import numpy as np
import re
from datetime import datetime
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from sklearn.preprocessing import OneHotEncoder
from sklearn.impute import SimpleImputer
import time
import warnings
warnings.filterwarnings("ignore")
```

```
[2]: from google.colab import files
uploaded = files.upload()
```

<IPython.core.display.HTML object>

Saving mudah-apartment-kl-selangor 3.csv to mudah-apartment-kl-selangor 3.csv

```
[3]: # Load raw dataset
df = pd.read_csv("mudah-apartment-kl-selangor 3.csv")
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19991 entries, 0 to 19990
Data columns (total 14 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   ads_id                                19991 non-null  int64
1   prop_name                             19043 non-null  object
2   completion_year                       10806 non-null  float64
3   monthly_rent                          19989 non-null  object
4   location                              19991 non-null  object
5   property_type                         19991 non-null  object
6   rooms                                 19985 non-null  object
7   parking                               14289 non-null  float64
8   bathroom                             19985 non-null  float64
9   size                                  19991 non-null  object
10  furnished                             19986 non-null  object
11  facilities                             17782 non-null  object
12  additional_facilities                 14043 non-null  object
13  region                                19991 non-null  object
dtypes: float64(3), int64(1), object(10)
memory usage: 2.1+ MB

```

```
[4]: df.head()
```

```

[4]:      ads_id      prop_name  completion_year \
0  100323185  The Hipster @ Taman Desa      2022.0
1  100203973      Segar Courts                NaN
2  100323128  Pangsapuri Teratak Muhibbah 2      NaN
3  100191767  Sentul Point Suite Apartment      2020.0
4   97022692      Arte Mont Kiara                NaN

      monthly_rent      location  property_type  rooms \
0  RM 4 200 per month  Kuala Lumpur - Taman Desa  Condominium    5
1  RM 2 300 per month      Kuala Lumpur - Cheras  Condominium    3
2  RM 1 000 per month  Kuala Lumpur - Taman Desa  Apartment     3
3  RM 1 700 per month      Kuala Lumpur - Sentul  Apartment     2
4  RM 1 299 per month  Kuala Lumpur - Mont Kiara  Service Residence  1

      parking  bathroom      size      furnished \
0        2.0        6.0  1842 sq.ft.  Fully Furnished
1        1.0        2.0  1170 sq.ft.  Partially Furnished
2        NaN        2.0   650 sq.ft.  Fully Furnished
3        1.0        2.0   743 sq.ft.  Partially Furnished
4        1.0        1.0   494 sq.ft.  Not Furnished

      facilities \
0  Minimart, Gymnasium, Security, Playground, Swi...

```

```

1 Playground, Parking, Barbeque area, Security, ...
2     Minimart, Jogging Track, Lift, Swimming Pool
3 Parking, Playground, Swimming Pool, Squash Cou...
4 Parking, Security, Lift, Swimming Pool, Playgr...

          additional_facilities      region
0     Air-Cond, Cooking Allowed, Washing Machine  Kuala Lumpur
1     Air-Cond, Cooking Allowed, Near KTM/LRT  Kuala Lumpur
2                                     NaN  Kuala Lumpur
3     Cooking Allowed, Near KTM/LRT, Washing Machine  Kuala Lumpur
4                                     Air-Cond  Kuala Lumpur

```

It can be observed that the raw dataset is quite messy. Several issues are identified as follows:

ads_id and prop_name may be irrelevant or non-informative variables.

completion_year and parking contain a large number of missing (NA) values.

monthly_rent is incorrectly read as a character (string) due to the presence of “RM” and “per month” in the entries.

location is a string-type variable and requires further preprocessing.

property_type is also a string and needs to be transformed appropriately.

rooms and bathroom appear to be fine at first glance.

size is incorrectly treated as a string due to the inclusion of “sq.ft”.

furnished is a string-type variable that also needs to be converted.

facilities and additional_facilities seem unstructured and require additional processing.

region is a string-type variable that also requires transformation.

Data Pre-processing

Firstly, check and delete the duplicated data from the raw dataset.

```

[5]: # Removing 'RM' from monthly rent and rename the column
df['monthly_rent'] = df['monthly_rent'].str.replace(r'RM|per month|', '', regex=True)
df['monthly_rent'] = df['monthly_rent'].str.replace(r'\s+', '', regex=True)
df['monthly_rent'] = pd.to_numeric(df['monthly_rent'], errors='coerce')

df.rename(columns={'monthly_rent': 'monthly_rent_rm'}, inplace=True)

```

For monthly_rent, remove the RM, per month, transform it as numeric and rename the column as monthly_rent_rm.

```

[6]: # Dropping 'sq.ft' from size
df['size'] = df['size'].str.extract(r'(.*) (?=\s*squ)', expand=False)
df['size'] = pd.to_numeric(df['size'], errors='coerce')

```

```
df.rename(columns={'size': 'size_sqft'}, inplace=True)
```

For size, remove the sq.ft, transform it as numeric and and rename the column as size_sqft.

```
[7]: # This step standardizes the location column by removing any additional
      ↪ neighborhood details after the main region name (e.g., "Kuala Lumpur - Taman
      ↪ Desa" becomes "Kuala Lumpur").
      # This helps in consistent grouping and regional analysis

df['location_cleaned'] = df['location'].str.extract(r'^(Kuala
      ↪ Lumpur|Selangor)', expand=False)

df['location_encoded'] = df['location_cleaned'].map({'Kuala Lumpur': 1,
      ↪ 'Selangor': 2})
```

```
[8]: df.head()
```

```
[8]:      ads_id      prop_name  completion_year  monthly_rent_rm \
0  100323185  The Hipster @ Taman Desa      2022.0      4200.0
1  100203973      Segar Courts              NaN      2300.0
2  100323128  Pangsapuri Teratak Muhibbah 2      NaN      1000.0
3  100191767  Sentul Point Suite Apartment      2020.0      1700.0
4   97022692      Arte Mont Kiara              NaN      1299.0
```

```
      location      property_type  rooms  parking  bathroom \
0  Kuala Lumpur - Taman Desa      Condominium      5      2.0      6.0
1    Kuala Lumpur - Cheras      Condominium      3      1.0      2.0
2  Kuala Lumpur - Taman Desa      Apartment      3      NaN      2.0
3    Kuala Lumpur - Sentul      Apartment      2      1.0      2.0
4  Kuala Lumpur - Mont Kiara  Service Residence      1      1.0      1.0
```

```
      size_sqft      furnished \
0      1842      Fully Furnished
1      1170  Partially Furnished
2       650      Fully Furnished
3       743  Partially Furnished
4       494      Not Furnished
```

```
      facilities \
0  Minimart, Gymnasium, Security, Playground, Swi...
1  Playground, Parking, Barbeque area, Security, ...
2      Minimart, Jogging Track, Lift, Swimming Pool
3  Parking, Playground, Swimming Pool, Squash Cou...
4  Parking, Security, Lift, Swimming Pool, Playgr...
```

	additional_facilities	region \
0	Air-Cond, Cooking Allowed, Washing Machine	Kuala Lumpur
1	Air-Cond, Cooking Allowed, Near KTM/LRT	Kuala Lumpur
2	NaN	Kuala Lumpur
3	Cooking Allowed, Near KTM/LRT, Washing Machine	Kuala Lumpur
4	Air-Cond	Kuala Lumpur

	location_cleaned	location_encoded
0	Kuala Lumpur	1
1	Kuala Lumpur	1
2	Kuala Lumpur	1
3	Kuala Lumpur	1
4	Kuala Lumpur	1

```
[9]: # In this step, I create a new variable nearby_ktm_lrt to indicate whether a
      ↪property is located near public transportation (KTM or LRT).
      # I extract this information from the additional_facilities column, which
      ↪contains various facility details in text format.
      # If the text includes "Near KTM/LRT", we assign a value of 1; otherwise, we
      ↪assign 0.
      # This binary variable makes it easier to analyze how proximity to public
      ↪transit affects monthly rent.

      #KTM (Keretapi Tanah Melayu) and LRT (Light Rail Transit) are two major
      ↪rail-based public transportation systems in Malaysia:
      #KTM is a commuter train system that connects suburban and outskirt areas to
      ↪the city center.
      #LRT is an urban light rail system that serves densely populated areas within
      ↪the city.
      #Being close to a KTM or LRT station typically adds value to a rental property
      ↪because it provides tenants with easy and cost-effective access to
      ↪transportation.
      df['nearby_ktm_lrt'] = df['additional_facilities'].astype(str).str.
      ↪contains('Near KTM/LRT', case=False, na=False).astype(int)

      print(df[['additional_facilities', 'nearby_ktm_lrt']].head(10))
```

	additional_facilities	nearby_ktm_lrt
0	Air-Cond, Cooking Allowed, Washing Machine	0
1	Air-Cond, Cooking Allowed, Near KTM/LRT	1
2	NaN	0
3	Cooking Allowed, Near KTM/LRT, Washing Machine	1
4	Air-Cond	0
5	Cooking Allowed, Near KTM/LRT	1
6	Air-Cond, Cooking Allowed, Near KTM/LRT, Washi...	1
7	Air-Cond, Cooking Allowed, Near KTM/LRT, Washi...	1
8	Air-Cond, Cooking Allowed, Washing Machine, Ne...	1

For location, remove the province name, such as KL and Selangor.

Further transformation will process after visualization.

```
[10]: # Facilities and additional_facilities processing
df['fac_count'] = df['facilities'].str.count(',') + 1
df['add_fac_count'] = df['additional_facilities'].str.count(',') + 1

def extract_near_ktm_lrt(text):
    if isinstance(text, str):
        keywords = ['ktm', 'lrt', 'mrt', 'monorail']
        return any(kw.lower() in text.lower() for kw in keywords)
    return False

df['nearby_ktm_lrt'] = df['additional_facilities'].apply(extract_near_ktm_lrt)
```

I count the number of facilities listed in both `facilities` and `additional_facilities` columns by counting the number of commas. Then, I create two new columns: `fac_count` and `add_fac_count` to store the respective counts.

```
[11]: # drop ads_id, prop_name, facilities, additional_facilities, location
df = df.drop(columns=['ads_id', 'prop_name', 'facilities',
    ↪ 'additional_facilities', 'location'])

df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19991 entries, 0 to 19990
Data columns (total 14 columns):
#   Column                Non-Null Count  Dtype
---  -
0   completion_year        10806 non-null  float64
1   monthly_rent_rm        19989 non-null  float64
2   property_type          19991 non-null  object
3   rooms                  19985 non-null  object
4   parking                 14289 non-null  float64
5   bathroom               19985 non-null  float64
6   size_sqft              19991 non-null  int64
7   furnished              19986 non-null  object
8   region                 19991 non-null  object
9   location_cleaned       19991 non-null  object
10  location_encoded       19991 non-null  int64
11  nearby_ktm_lrt         19991 non-null  bool
12  fac_count              17782 non-null  float64
13  add_fac_count          14043 non-null  float64
dtypes: bool(1), float64(6), int64(2), object(5)
memory usage: 2.0+ MB
```

```
[12]: df.head()
```

```
[12]: completion_year  monthly_rent_rm    property_type rooms  parking \
0          2022.0          4200.0      Condominium     5      2.0
1           NaN          2300.0      Condominium     3      1.0
2           NaN          1000.0      Apartment       3      NaN
3          2020.0          1700.0      Apartment     2      1.0
4           NaN          1299.0  Service Residence     1      1.0

bathroom  size_sqft    furnished    region location_cleaned \
0         6.0      1842    Fully Furnished  Kuala Lumpur    Kuala Lumpur
1         2.0      1170  Partially Furnished  Kuala Lumpur    Kuala Lumpur
2         2.0       650    Fully Furnished  Kuala Lumpur    Kuala Lumpur
3         2.0       743  Partially Furnished  Kuala Lumpur    Kuala Lumpur
4         1.0       494    Not Furnished   Kuala Lumpur    Kuala Lumpur

location_encoded  nearby_ktm_lrt  fac_count  add_fac_count
0                1             False      10.0           3.0
1                1              True       9.0           3.0
2                1             False       4.0           NaN
3                1              True       8.0           3.0
4                1             False      11.0           1.0
```

Dropped the 4 useless columns ads_id, prop_name, facilities, additional_facilities.

Visualization

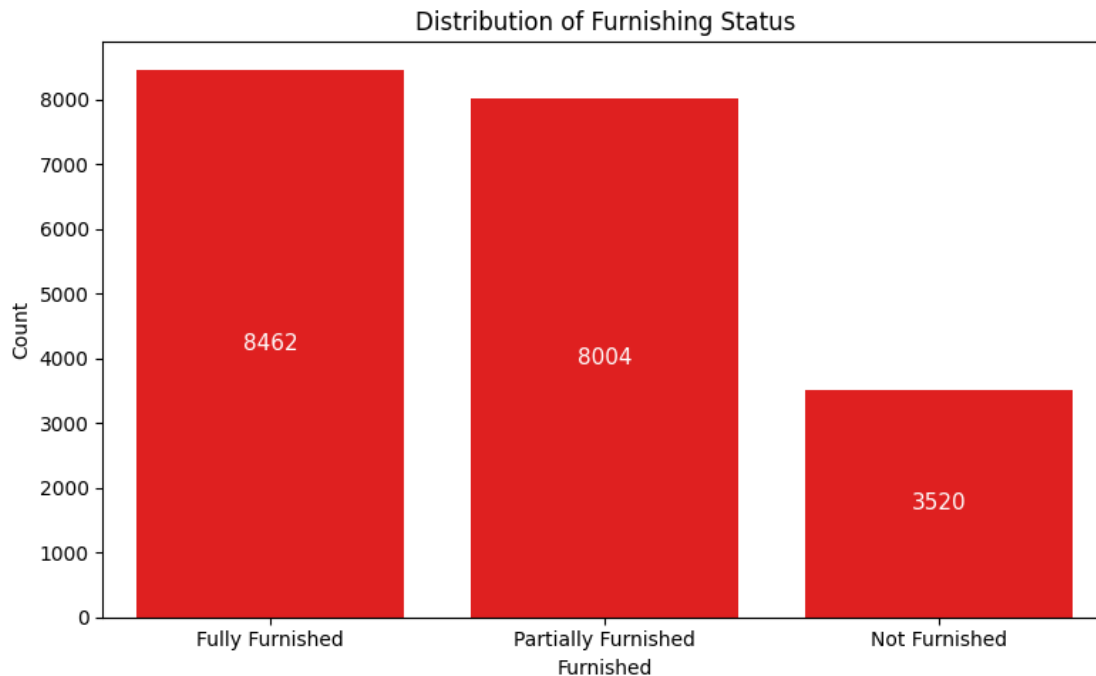
```
[ ]: # Bar plot_Distribution of Furnishing Status

plt.figure(figsize=(8, 5))
ax = sns.countplot(data=df, x='furnished', color='red')

for p in ax.patches:
    count = int(p.get_height())
    ax.annotate(f'{count}', (p.get_x() + p.get_width() / 2., count / 2),
                ha='center', va='center', fontsize=11, color='white')

ax.legend_.remove() if ax.legend_ else None

ax.set_title('Distribution of Furnishing Status')
plt.xlabel('Furnished')
plt.ylabel('Count')
plt.tight_layout()
plt.show()
```

```
[ ]: plt.figure(figsize=(8, 5))

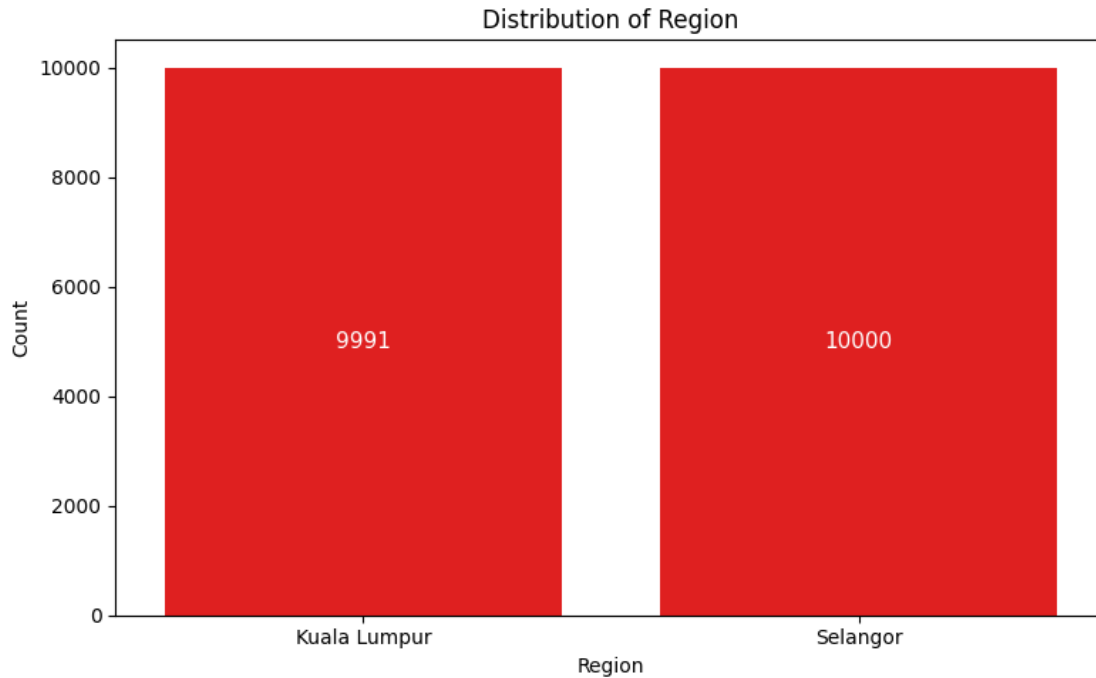
ax = sns.countplot(data=df, x='region', color='red')

for p in ax.patches:
    count = int(p.get_height())
    ax.annotate(f'{count}',
                (p.get_x() + p.get_width() / 2., count / 2),
                ha='center', va='center',
                fontsize=11, color='white')

if ax.get_legend() is not None:
    ax.get_legend().remove()

ax.set_title('Distribution of Region')
plt.xlabel('Region')
plt.ylabel('Count')

plt.tight_layout()
plt.show()
```



```
[ ]: df['completion_year'] = df['completion_year'].dropna().astype(int)

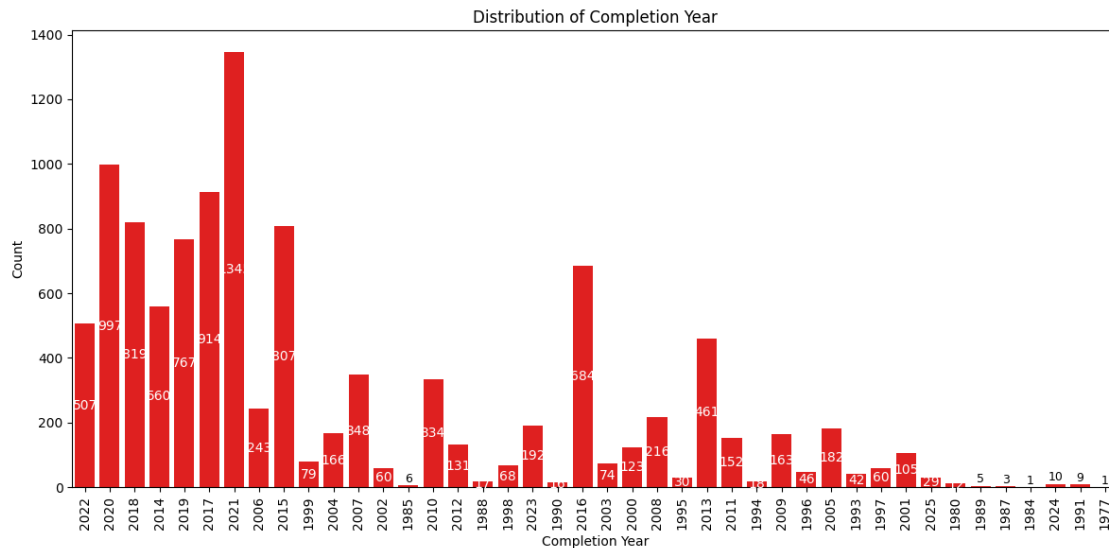
plt.figure(figsize=(12, 6))
ax = sns.countplot(data=df_completion, x='completion_year', color='red')

for p in ax.patches:
    count = int(p.get_height())
    if count > 10:
        ax.annotate(f'{count}',
                    (p.get_x() + p.get_width() / 2., count / 2),
                    ha='center', va='center',
                    fontsize=10, color='white')
    else:
        ax.annotate(f'{count}',
                    (p.get_x() + p.get_width() / 2., count + 1),
                    ha='center', va='bottom',
                    fontsize=9, color='black')

if ax.get_legend() is not None:
    ax.get_legend().remove()

ax.set_title('Distribution of Completion Year')
plt.xlabel('Completion Year')
plt.ylabel('Count')
plt.xticks(rotation=90)
```

```
plt.tight_layout()
plt.show()
```



```
[ ]: df['bathroom'] = pd.to_numeric(df['bathroom'], errors='coerce')
df_bathroom = df[df['bathroom'].notna()].copy()
df_bathroom['bathroom'] = df_bathroom['bathroom'].astype(int).astype(str)

plt.figure(figsize=(10, 5))
ax = sns.countplot(data=df_bathroom, x='bathroom', color='red')

for p in ax.patches:
    count = int(p.get_height())
    if count > 10:
        ax.annotate(f'{count}',
                    (p.get_x() + p.get_width() / 2., count / 2),
                    ha='center', va='center',
                    fontsize=10, color='white')
    else:
        ax.annotate(f'{count}',
                    (p.get_x() + p.get_width() / 2., count + 1),
                    ha='center', va='bottom',
                    fontsize=9, color='black')

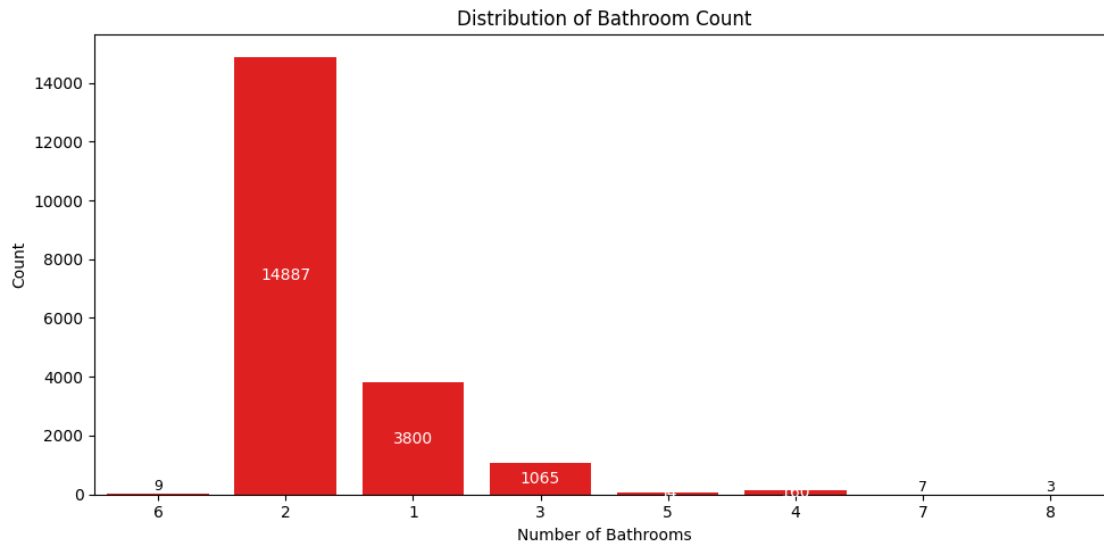
ax.set_title('Distribution of Bathroom Count')
plt.xlabel('Number of Bathrooms')
plt.ylabel('Count')
plt.tight_layout()
```

```

if ax.get_legend() is not None:
    ax.get_legend().remove()

plt.show()

```



- “Fully Furnished” has the highest count among furnishing statuses, and there are 5 missing values in this column.
- Kuala Lumpur and Selangor have nearly the same number of rental listings.
- More rental listings appeared in 2021 than in any other year.
- Most rental properties have 2 bathrooms, and there are 3 listings with 8 bathrooms.

```

[ ]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df_bar = df[['rooms', 'bathroom', 'monthly_rent_rm']].dropna()

df_bar['rooms'] = pd.to_numeric(df_bar['rooms'], errors='coerce')
df_bar['bathroom'] = pd.to_numeric(df_bar['bathroom'], errors='coerce')

df_bar = df_bar.dropna(subset=['rooms', 'bathroom'])

df_bar['rooms'] = df_bar['rooms'].astype(int)
df_bar['bathroom'] = df_bar['bathroom'].astype(int)

room_order = sorted(df_bar['rooms'].unique())
df_bar['rooms'] = pd.Categorical(df_bar['rooms'], categories=room_order,
    ↪ordered=True)

```

```

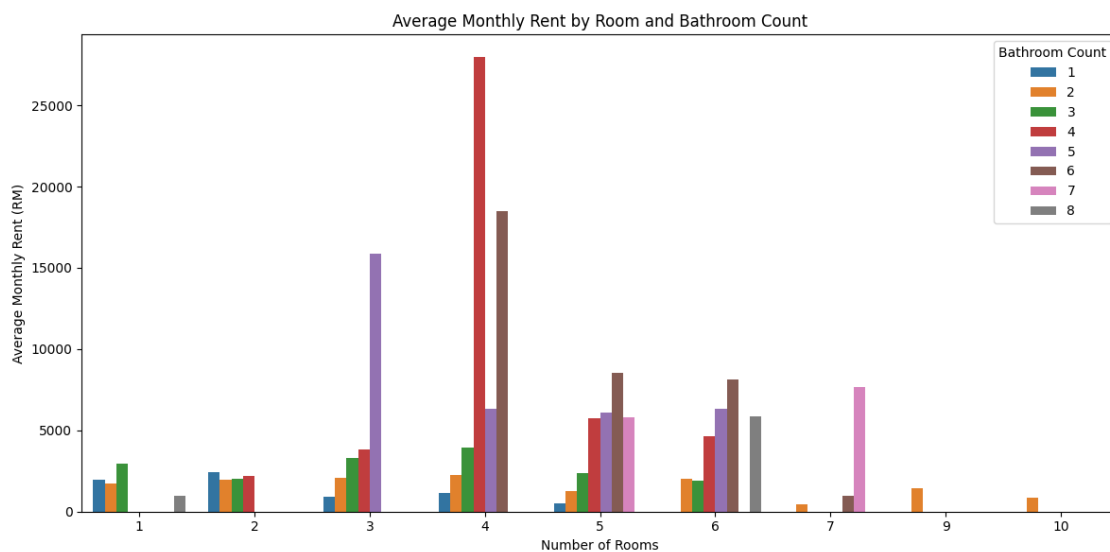
df_grouped = df_bar.groupby(['rooms', 'bathroom'])['monthly_rent_rm'].mean().
    ↪reset_index()

df_grouped['bathroom'] = df_grouped['bathroom'].astype(str)

plt.figure(figsize=(12, 6))
ax = sns.barplot(data=df_grouped, x='rooms', y='monthly_rent_rm',
    ↪hue='bathroom')

ax.set_title('Average Monthly Rent by Room and Bathroom Count')
ax.set_xlabel('Number of Rooms')
ax.set_ylabel('Average Monthly Rent (RM)')
plt.legend(title='Bathroom Count')
plt.tight_layout()
plt.show()

```



Based on the chart, houses with 4 rooms and 4–6 bathrooms have the highest average monthly rent, indicating that larger units with more bathrooms tend to command higher rental prices.

```

[ ]: # Parallel Coordinate Plot
     # Normalization is performed to improve the visualization of the parallel
     ↪coordinates plot.

```

```

[ ]: import pandas as pd
     from sklearn.preprocessing import MinMaxScaler
     import plotly.graph_objects as go

     df_encoded = df.copy()

```

```

df_encoded['property_type'] = df_encoded['property_type'].astype('category').
    ↪cat.codes

df_encoded['rooms'] = pd.to_numeric(df_encoded['rooms'].replace({'More than 10':
    ↪ 11}), errors='coerce')
df_encoded['parking'] = pd.to_numeric(df_encoded['parking'], errors='coerce')

df_norm = df_encoded[['property_type', 'rooms', 'parking', 'monthly_rent_rm']].
    ↪dropna()
df_norm = df_norm[df_norm['monthly_rent_rm'] < 20000]

scaler = MinMaxScaler()
df_norm[['property_type', 'rooms', 'parking']] = scaler.
    ↪fit_transform(df_norm[['property_type', 'rooms', 'parking']])

fig = go.Figure(data=go.Parcoords(
    line=dict(
        color=df_norm['monthly_rent_rm'],
        colorscale='Turbo',
        showscale=True,
        cmin=df_norm['monthly_rent_rm'].min(),
        cmax=df_norm['monthly_rent_rm'].max(),
        colorbar=dict(title='Monthly Rent (RM)')
    ),
    dimensions=[
        dict(label='Property Type', values=df_norm['property_type']),
        dict(label='Rooms', values=df_norm['rooms']),
        dict(label='Parking', values=df_norm['parking']),
    ]
))

fig.update_layout(title='Parallel Coordinates Plot of Property Features',
    ↪font=dict(size=14))
fig.show()

```

The parallel coordinates plot is used to visualize the relationships between normalized variables: `property_type`, `rooms`, and `parking`, colored by `monthly_rent_rm`. Each line represents one property, connecting values of different features.

From the plot, we can observe that properties with more rooms and more parking tend to have higher rental prices (represented by redder lines), whereas smaller units with fewer rooms and parking are associated with lower rental prices (bluer lines).

However, due to uncleaned data and outliers, the plot may appear cluttered and patterns may be less obvious. Additional cleaning or filtering may help highlight stronger trends.

```
[ ]: import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

df['rooms'] = pd.to_numeric(df['rooms'].replace({'More than 10': 11}),
    errors='coerce')
df['parking'] = pd.to_numeric(df['parking'], errors='coerce')
df['bathroom'] = pd.to_numeric(df['bathroom'], errors='coerce')
df['monthly_rent_rm'] = pd.to_numeric(df['monthly_rent_rm'], errors='coerce')

variables = ['bathroom', 'parking', 'monthly_rent_rm', 'rooms']

fig, axes = plt.subplots(nrows=4, ncols=2, figsize=(14, 24))
axes = axes.flatten()

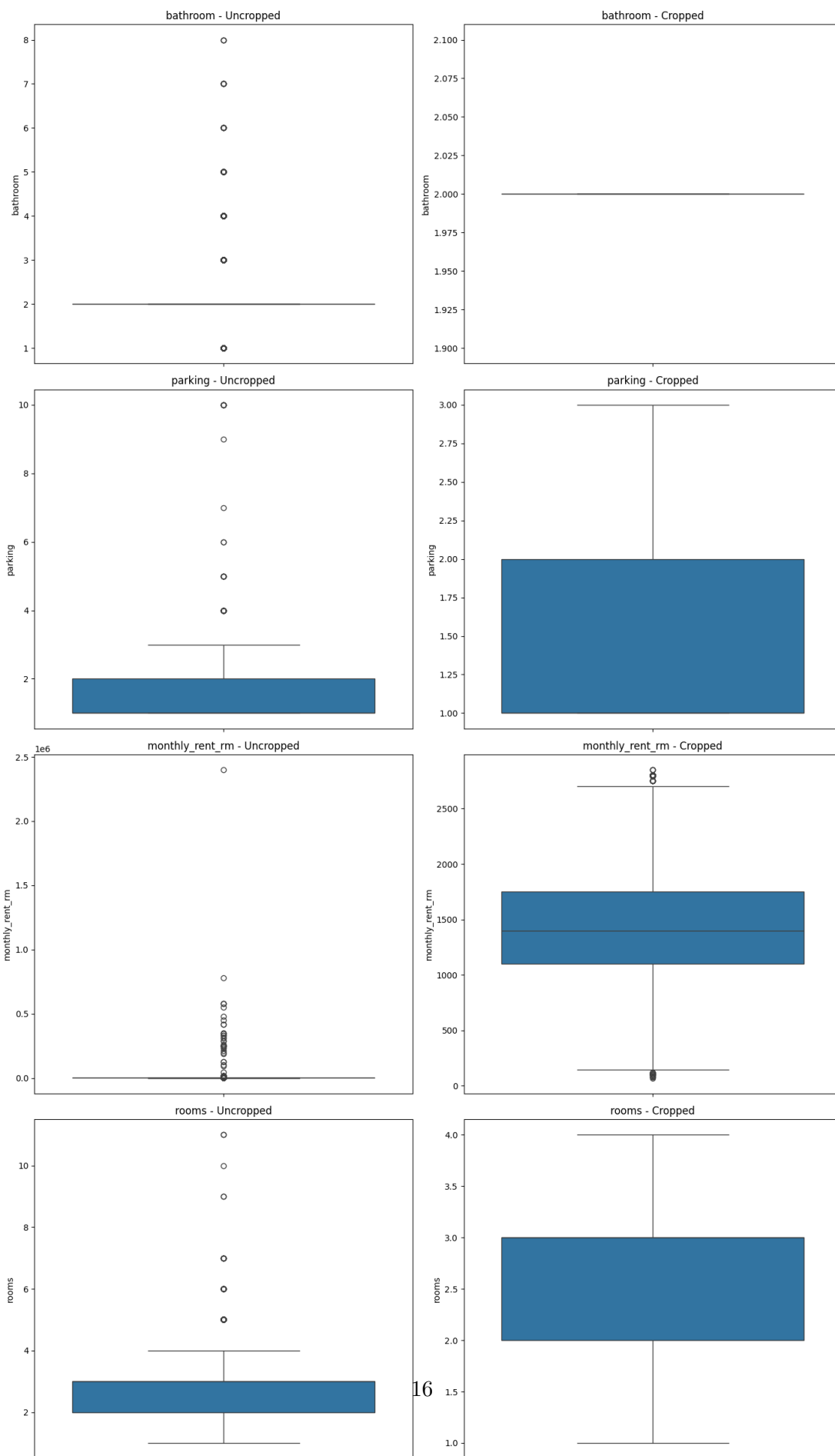
for i, var in enumerate(variables):
    series = df[var].dropna()

    sns.boxplot(y=series, ax=axes[2*i])
    axes[2*i].set_title(f'{var} - Uncropped')

    q1 = series.quantile(0.25)
    q3 = series.quantile(0.75)
    iqr = q3 - q1
    lower = q1 - 1.5 * iqr
    upper = q3 + 1.5 * iqr
    cropped = series[(series >= lower) & (series <= upper)]

    sns.boxplot(y=cropped, ax=axes[2*i + 1])
    axes[2*i + 1].set_title(f'{var} - Cropped')

plt.tight_layout()
plt.show()
```



We can observed significant outliers in rooms, parking, and bathrooms.

After applying IQR-based cropping, most distributions appeared more consistent and reliable for downstream modeling.

```
[13]: # Pie Chart for nominal variables
```

```
[14]: #This is an interactive pie chart. You can click on each color segment to
      ↪highlight it and view its proportion more clearly.
import plotly.express as px

fig = px.pie(df, names='furnished', title='Distribution of Furnished Status')
fig.show()
```

```
[15]: #This is an interactive pie chart. You can click on each color segment to
      ↪highlight it and view its proportion more clearly.

def create_pie_chart_plotly(df, column, title=None):
    fig = px.pie(df, names=column, title=title if title else f'Distribution of_{column}')
    fig.show()

create_pie_chart_plotly(df, "property_type", title="Distribution of Property
      ↪Types")
```

```
[16]: #This is an interactive pie chart. You can click on each color segment to
      ↪highlight it and view its proportion more clearly.

def create_pie_chart_plotly(df, column, title=None):
    fig = px.pie(df, names=column, title=title if title else f'Distribution of_{column}')
    fig.show()

create_pie_chart_plotly(df, "region", title="Distribution of Region")
```

I used pie charts to provide an overview of the distribution of nominal variables. As shown, all nominal variables exhibit fairly balanced distributions, except for `property_type`. For `property_type`, more than 95% of the data falls under Apartment, Condominium, and Service Residence.”

```
[ ]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

numeric_vars = ['completion_year', 'rooms', 'parking', 'bathroom']
```

```

for col in numeric_vars:
    df[col] = pd.to_numeric(df[col], errors='coerce')

fig, axes = plt.subplots(nrows=2, ncols=2, figsize=(16, 18))
axes = axes.flatten()

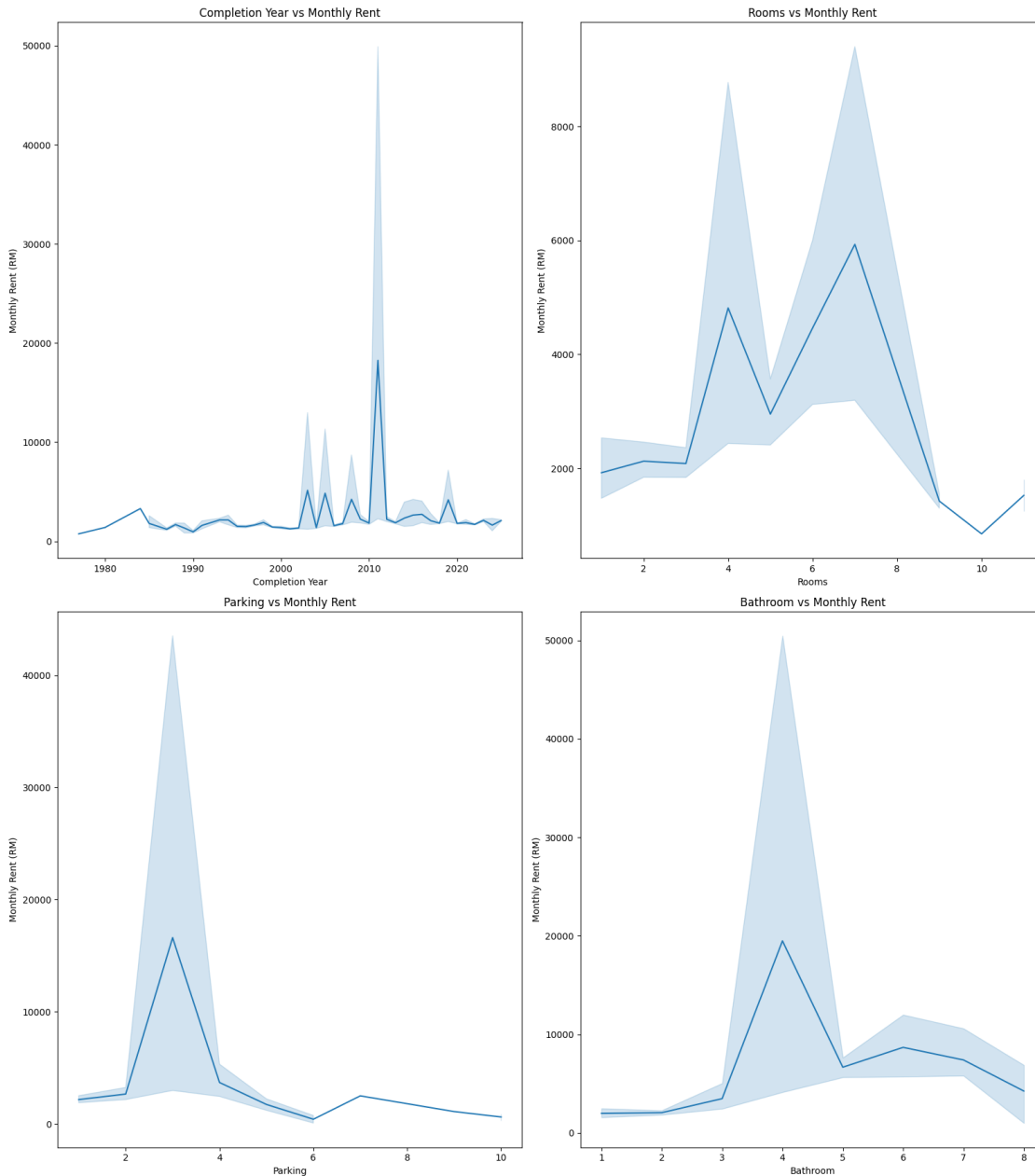
titles = [f'{v.replace("_", " ").title()} vs Monthly Rent' for v in
    ↪numeric_vars]

for i, var in enumerate(numeric_vars):
    temp_df = df[[var, 'monthly_rent_rm']].dropna()
    if not temp_df.empty:
        sns.lineplot(data=temp_df, x=var, y='monthly_rent_rm', ax=axes[i])
        axes[i].set_title(titles[i])
        axes[i].set_xlabel(var.replace('_', ' ').title())
        axes[i].set_ylabel('Monthly Rent (RM)')
    else:
        axes[i].text(0.5, 0.5, f"No valid data for {var}", ha='center')
        axes[i].axis('off')

if len(numeric_vars) < len(axes):
    axes[-1].axis('off')

plt.tight_layout()
plt.show()

```



#1. Completion Year vs Monthly Rent Newer properties (post-2000) show more stable rental prices around RM 800, whereas older buildings have more variation and some extremely low rents. This might be due to deterioration or smaller sample size in earlier years.

#2. Rooms vs Monthly Rent Generally, properties with 1–3 rooms have relatively high average rents. However, rent surprisingly decreases for 4–6 rooms, possibly due to shared housing. A sharp increase at 10 rooms may indicate luxury or commercial properties.

#3. Parking vs Monthly Rent

Units with 1–2 parking spaces have higher and more stable rents. Units with 6 or more parking

spaces show wide variability and lower average rent, which could be influenced by outliers or niche property types.

#4. Bathroom vs Monthly Rent

Rent increases with bathroom count up to 2, but adding more bathrooms beyond that does not guarantee higher rent. The rent for units with 4 bathrooms is actually lower, suggesting diminishing returns or data sparsity.

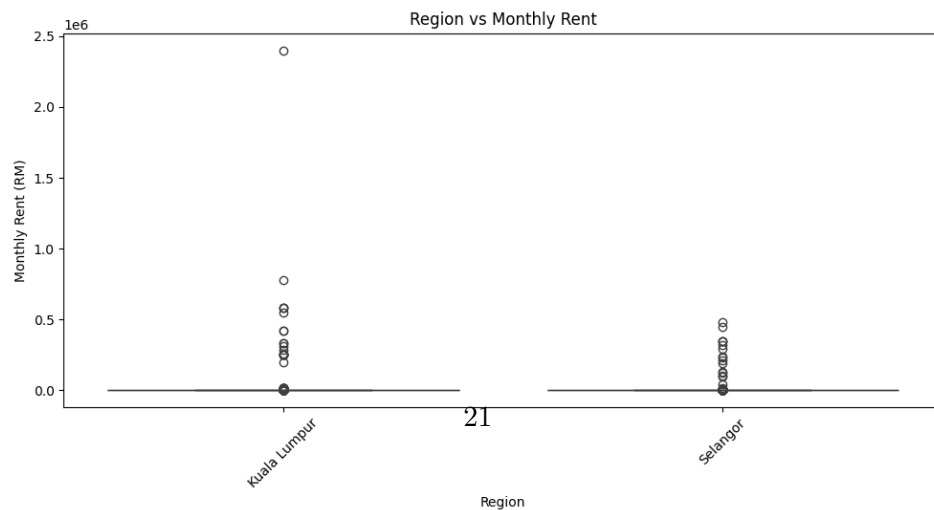
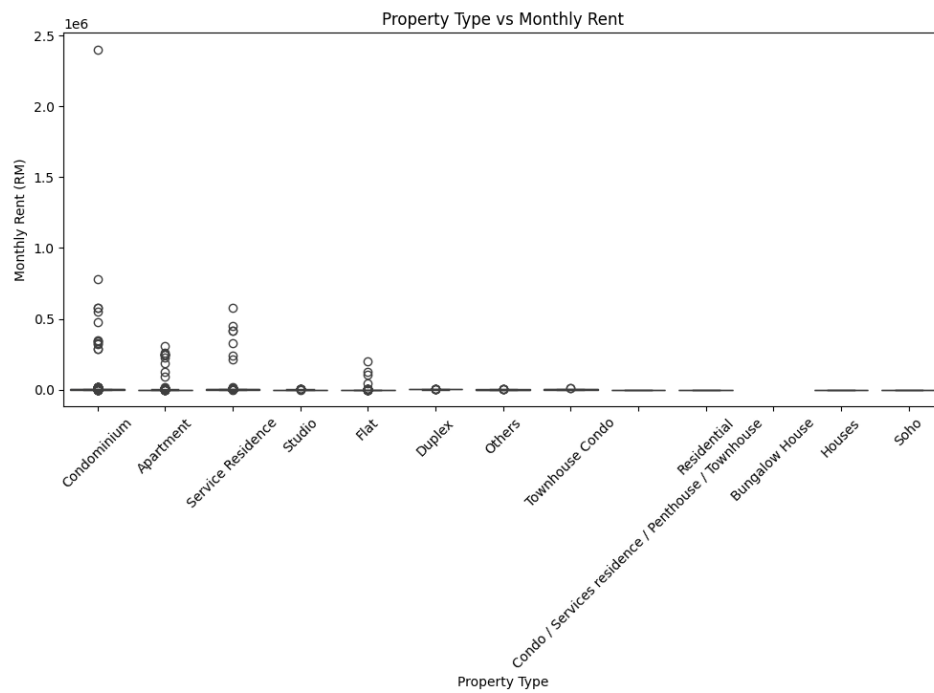
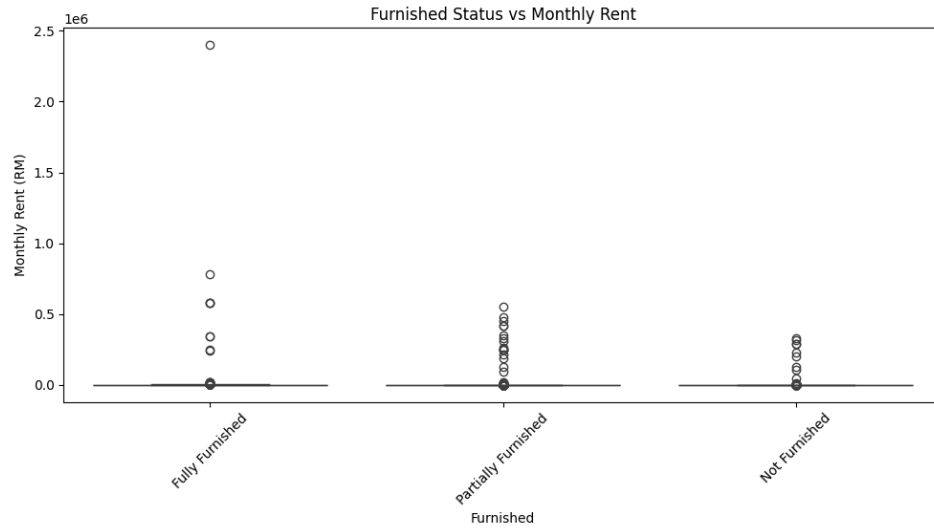
```
[ ]: import matplotlib.pyplot as plt
import seaborn as sns

fig, axes = plt.subplots(nrows=3, ncols=1, figsize=(10, 20))
axes = axes.flatten()

cat_vars = ['furnished', 'property_type', 'region']
titles = ['Furnished Status vs Monthly Rent',
          'Property Type vs Monthly Rent',
          'Region vs Monthly Rent']

for i, var in enumerate(cat_vars):
    sns.boxplot(data=df, x=var, y='monthly_rent_rm', ax=axes[i])
    axes[i].set_title(titles[i])
    axes[i].set_xlabel(var.replace('_', ' ').title())
    axes[i].set_ylabel('Monthly Rent (RM)')
    axes[i].tick_params(axis='x', rotation=45)

plt.tight_layout()
plt.show()
```



1 1. Furnished Status vs Monthly Rent

Fully furnished and partially furnished properties tend to have higher median rental prices compared to non-furnished units. However, all categories show a wide range of rent values with visible outliers, indicating rental price variation within each furnishing group.

#2. Property Type vs Monthly Rent Apartments, condominiums, and service residences dominate the dataset and show relatively similar rent distributions, though service residences have a slightly higher median. Less common property types such as “Studio,” “Flat,” and “Duplex” also show consistent distributions. Some rare types (e.g., “Soho,” “Residential”) appear to have either very few listings or inconsistent rent values, resulting in narrow or flat boxplots.

#3. Region vs Monthly Rent Properties in Kuala Lumpur tend to have slightly higher rent compared to those in Selangor. Both regions exhibit a similar spread in rental prices and contain lower-end outliers. The median rental price is slightly above RM 800 for both regions.

##Data cleaning after visualization

```
[18]: df.head()
```

```
[18]: completion_year  monthly_rent_rm    property_type rooms  parking \
0          2022.0         4200.0      Condominium     5      2.0
1           NaN         2300.0      Condominium     3      1.0
2           NaN         1000.0      Apartment       3      NaN
3          2020.0         1700.0      Apartment     2      1.0
4           NaN         1299.0  Service Residence     1      1.0

    bathroom  size_sqft    furnished    region location_cleaned \
0         6.0      1842  Fully Furnished  Kuala Lumpur    Kuala Lumpur
1         2.0      1170  Partially Furnished  Kuala Lumpur    Kuala Lumpur
2         2.0       650  Fully Furnished  Kuala Lumpur    Kuala Lumpur
3         2.0       743  Partially Furnished  Kuala Lumpur    Kuala Lumpur
4         1.0       494    Not Furnished  Kuala Lumpur    Kuala Lumpur

    location_encoded  nearby_ktm_lrt  fac_count  add_fac_count
0                 1             False        10.0           3.0
1                 1              True         9.0           3.0
2                 1             False         4.0           NaN
3                 1              True         8.0           3.0
4                 1             False        11.0           1.0
```

```
[20]: # Noise/outliers processing
# After visualizing the data, I filtered the most appropriate ranges for the
↳ numeric variables to remove outliers and ensure data quality.
# Regarding property_type, I found that approximately 95% of the listings fall
↳ into three categories: "Apartment," "Condominium," and "Service Residence."
```

```

# Therefore, I retained only these types for further analysis.

df1 = df.copy()

df1['monthly_rent_rm'] = pd.to_numeric(df1['monthly_rent_rm'], errors='coerce')
df1['size_sqft'] = pd.to_numeric(df1['size_sqft'], errors='coerce')
df1['rooms'] = pd.to_numeric(df1['rooms'], errors='coerce')
df1['parking'] = pd.to_numeric(df1['parking'], errors='coerce')
df1['bathroom'] = pd.to_numeric(df1['bathroom'], errors='coerce')

df1 = df1[(df1['monthly_rent_rm'] > 400) & (df1['monthly_rent_rm'] < 3500)]
df1 = df1[(df1['size_sqft'] > 100) & (df1['size_sqft'] < 2000)]
df1 = df1[(df1['rooms'] > 0) & (df1['rooms'] < 4)]
df1 = df1[(df1['parking'] > 0) & (df1['parking'] < 4)]
df1 = df1[(df1['bathroom'] > 0) & (df1['bathroom'] < 4)]

df1 = df1[df1['property_type'].isin(["Apartment", "Condominium", "Service_
↪Residence"])]

```

```

[21]: # converts the furnished column from text categories (e.g., "Fully Furnished")
↪into numeric values.

furnish_map = {"Not Furnished": 1, "Partially Furnished": 2, "Fully Furnished":
↪3}
df1["furnished"] = df1["furnished"].map(furnish_map).round()

# encodes the region column into numeric form: "Kuala Lumpur" → 1 "Selangor" → 2
region_map = {"Kuala Lumpur": 1, "Selangor": 2}
df1["region"] = df1["region"].map(region_map).round()

# converts the nearby_ktm_lrt column from boolean values into numeric: True
↪(property is near KTM/LRT) → 1 False (not near KTM/LRT) → 2
ktm_map = {True: 1, False: 2}
df1["nearby_ktm_lrt"] = df1["nearby_ktm_lrt"].map(ktm_map).round()

#This step transforms the property_type column into numeric labels for modeling:
↪ "Condominium" → 1 "Apartment" → 2 "Service Residence" → 3
property_map = {"Condominium": 1, "Apartment": 2, "Service Residence": 3}
df1["property_type"] = df1["property_type"].map(property_map).round()

```

```

[22]: # count unique value for categorical data
is_categorical = df1.select_dtypes(include=['object'])

print("The number of unique values for categorical data:")
for col in is_categorical.columns:
    unique_count = df1[col].nunique()
    print(f"{col}: {unique_count} unique values")

```

The number of unique values for categorical data:
location_cleaned: 2 unique values

```
[23]: location_map = {"Kuala Lumpur": 1, "Selangor": 2}
df1["location_cleaned"] = df1["location_cleaned"].map(location_map)
```

```
[24]: # count again unique value for categorical data
is_categorical = df1.select_dtypes(include=['object'])

print("The number of unique values for categorical data:")
for col in is_categorical.columns:
    unique_count = df1[col].nunique()
    print(f"{col}: {unique_count} unique values")
```

The number of unique values for categorical data:

```
[25]: print(df1.shape)
print(df1.head())
```

(12308, 14)

	completion_year	monthly_rent_rm	property_type	rooms	parking	bathroom	\
1	NaN	2300.0	1	3.0	1.0	2.0	
3	2020.0	1700.0	2	2.0	1.0	2.0	
4	NaN	1299.0	3	1.0	1.0	1.0	
5	NaN	1500.0	2	3.0	1.0	2.0	
6	NaN	2900.0	3	3.0	2.0	2.0	

	size_sqft	furnished	region	location_cleaned	location_encoded	\
1	1170	2	1	1	1	
3	743	2	1	1	1	
4	494	1	1	1	1	
5	884	2	1	1	1	
6	982	3	1	1	1	

	nearby_ktm_lrt	fac_count	add_fac_count
1	1	9.0	3.0
3	1	8.0	3.0
4	2	11.0	1.0
5	1	6.0	2.0
6	1	10.0	4.0

```
[26]: # found missing values only in numerical columns, not in categorical ones like
      ↪ location.
      # To impute the missing values, I used the mode (most frequent value) rather
      ↪ than the mean, since variables like parking require integer values.

from sklearn.impute import SimpleImputer
```



```
import pandas as pd

num_cols = df1.select_dtypes(include=['float64', 'int64']).columns
X_num = df1[num_cols]

imputer = SimpleImputer(strategy='most_frequent')
X_num_imputed = pd.DataFrame(imputer.fit_transform(X_num), columns=num_cols)

print(X_num_imputed.isna().sum())
```

```
completion_year      0
monthly_rent_rm      0
property_type        0
rooms                0
parking              0
bathroom             0
size_sqft            0
furnished            0
region              0
location_cleaned     0
location_encoded     0
nearby_ktm_lrt       0
fac_count            0
add_fac_count        0
dtype: int64
```

1.1 Split the dataset: 70:15:15

```
[27]: # Before building the regression models, I set monthly_rent_rm as the target
      ↪ variable (y) and the remaining features as predictors (X).
      # Then, I split the dataset into training, validation, and testing sets: 70:15:
      ↪ 15
      from sklearn.model_selection import train_test_split
      from sklearn.preprocessing import StandardScaler

      X1 = df1.drop(columns=['monthly_rent_rm'])
      y1 = df1['monthly_rent_rm']

      scaler = StandardScaler()
      X_clean1 = scaler.fit_transform(X1)

      X_temp, X_test_val, y_temp, y_test_val = train_test_split(X_clean1, y1,
      ↪ test_size=0.30, random_state=123)
      X_val, X_test, y_val, y_test = train_test_split(X_test_val, y_test_val,
      ↪ test_size=0.5, random_state=123)

      print("Train set:", X_temp.shape)
```

```
print("Validation set:", X_val.shape)
print("Test set:", X_test.shape)
```

Train set: (8615, 13)

Validation set: (1846, 13)

Test set: (1847, 13)

Train linear regression model

```
[28]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import time
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

# Train Set
X_temp_df = pd.DataFrame(X_temp)
y_temp_series = pd.Series(y_temp).reset_index(drop=True)
train_df = pd.concat([X_temp_df, y_temp_series.rename("monthly_rent_rm")],
    ↪axis=1)
train_df_clean = train_df.dropna()

X_temp_clean = train_df_clean.drop(columns=["monthly_rent_rm"]).values
y_temp_clean = train_df_clean["monthly_rent_rm"].values

X_temp_clean_b = np.c_[np.ones((X_temp_clean.shape[0], 1)), X_temp_clean]

start_time = time.time()
theta = np.linalg.pinv(X_temp_clean_b) @ y_temp_clean
end_time = time.time()
print("Training time (Normal Equation): {:.2f} seconds".format(end_time -
    ↪start_time))
```

Training time (Normal Equation): 0.04 seconds

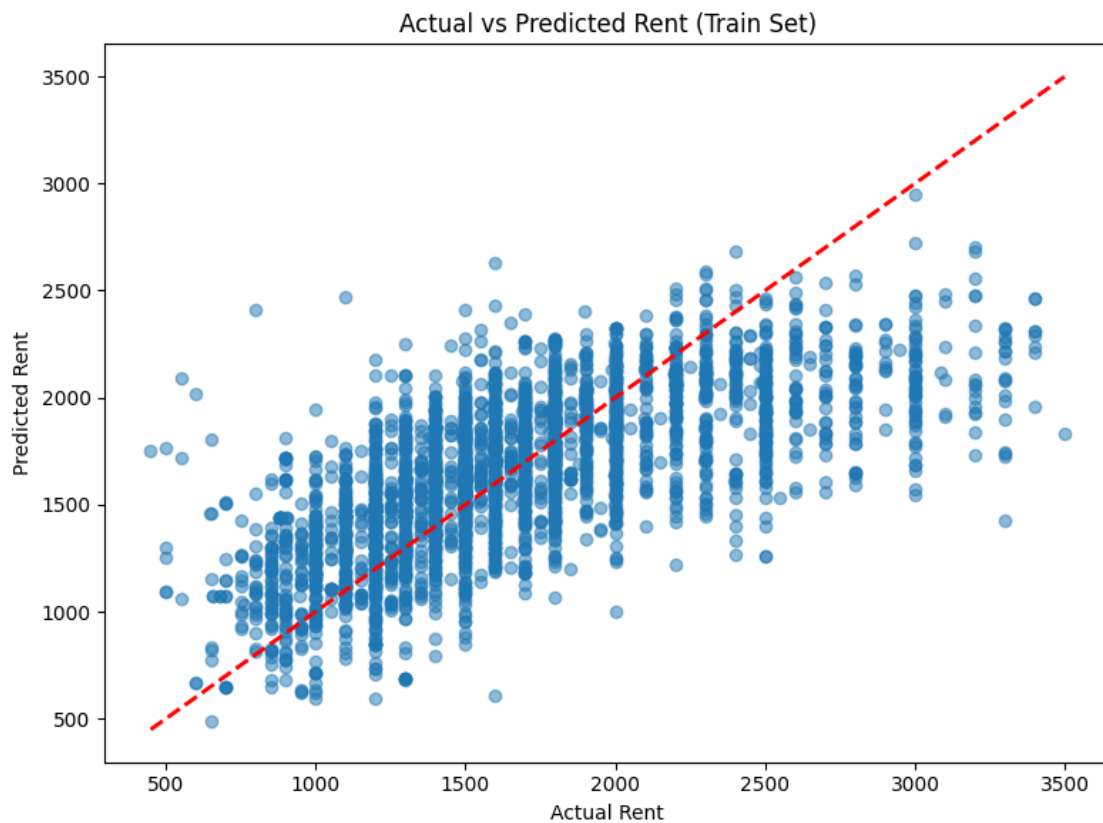
```
[29]: y_pred_train = X_temp_clean_b @ theta

mse = mean_squared_error(y_temp_clean, y_pred_train)
mae = mean_absolute_error(y_temp_clean, y_pred_train)
r2 = r2_score(y_temp_clean, y_pred_train)

print("Train Performance")
print(f"Train MSE: {mse:.2f}")
print(f"Train MAE: {mae:.2f}")
print(f"Train R²: {r2:.2f}")
```

Train Performance
Train MSE: 146830.82
Train MAE: 287.13
Train R^2 : 0.47

```
[30]: plt.figure(figsize=(8, 6))
plt.scatter(y_temp_clean, y_pred_train, alpha=0.5)
plt.plot([min(y_temp_clean), max(y_temp_clean)], [min(y_temp_clean),
↪max(y_temp_clean)],
         color='red', linestyle='--', linewidth=2)
plt.xlabel('Actual Rent')
plt.ylabel('Predicted Rent')
plt.title('Actual vs Predicted Rent (Train Set)')
plt.tight_layout()
plt.show()
```



```
[31]: # Test Set
X_test_df = pd.DataFrame(X_test)
y_test_series = pd.Series(y_test).reset_index(drop=True)
test_df = pd.concat([X_test_df, y_test_series.rename("monthly_rent_rm")],
↪axis=1)
```

```

test_df_clean = test_df.dropna()

X_test_clean = test_df_clean.drop(columns=["monthly_rent_rm"]).values
y_test_clean = test_df_clean["monthly_rent_rm"].values

X_test_clean_b = np.c_[np.ones((X_test_clean.shape[0], 1)), X_test_clean]

y_pred_test = X_test_clean_b @ theta

mse_test = mean_squared_error(y_test_clean, y_pred_test)
mae_test = mean_absolute_error(y_test_clean, y_pred_test)
r2_test = r2_score(y_test_clean, y_pred_test)

print("Test Set Performance")
print(f"Test MSE: {mse_test:.2f}")
print(f"Test MAE: {mae_test:.2f}")
print(f"Test R2: {r2_test:.2f}")

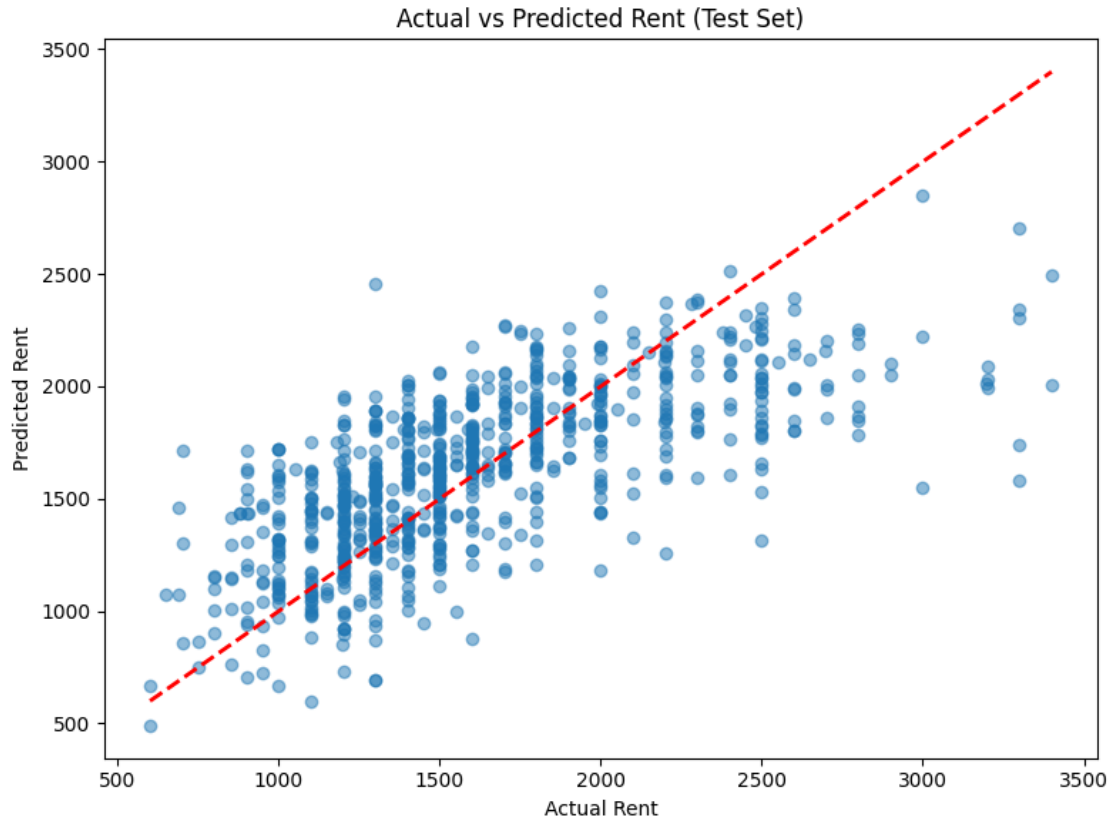
```

Test Set Performance
 Test MSE: 138270.30
 Test MAE: 282.56
 Test R²: 0.47

```

[32]: # Actual vs Predicted Rent (Test set) plot
plt.figure(figsize=(8, 6))
plt.scatter(y_test_clean, y_pred_test, alpha=0.5)
plt.plot([min(y_test_clean), max(y_test_clean)],
         [min(y_test_clean), max(y_test_clean)],
         color='red', linestyle='--', linewidth=2)
plt.xlabel('Actual Rent')
plt.ylabel('Predicted Rent')
plt.title('Actual vs Predicted Rent (Test Set)')
plt.tight_layout()
plt.show()

```



```
[33]: #Valid Set
X_val_df = pd.DataFrame(X_val)
y_val_series = pd.Series(y_val).reset_index(drop=True)
val_df = pd.concat([X_val_df, y_val_series.rename("monthly_rent_rm")], axis=1)
val_df_clean = val_df.dropna()

X_val_clean = val_df_clean.drop(columns=["monthly_rent_rm"]).values
y_val_clean = val_df_clean["monthly_rent_rm"].values

X_val_clean_b = np.c_[np.ones((X_val_clean.shape[0], 1)), X_val_clean]

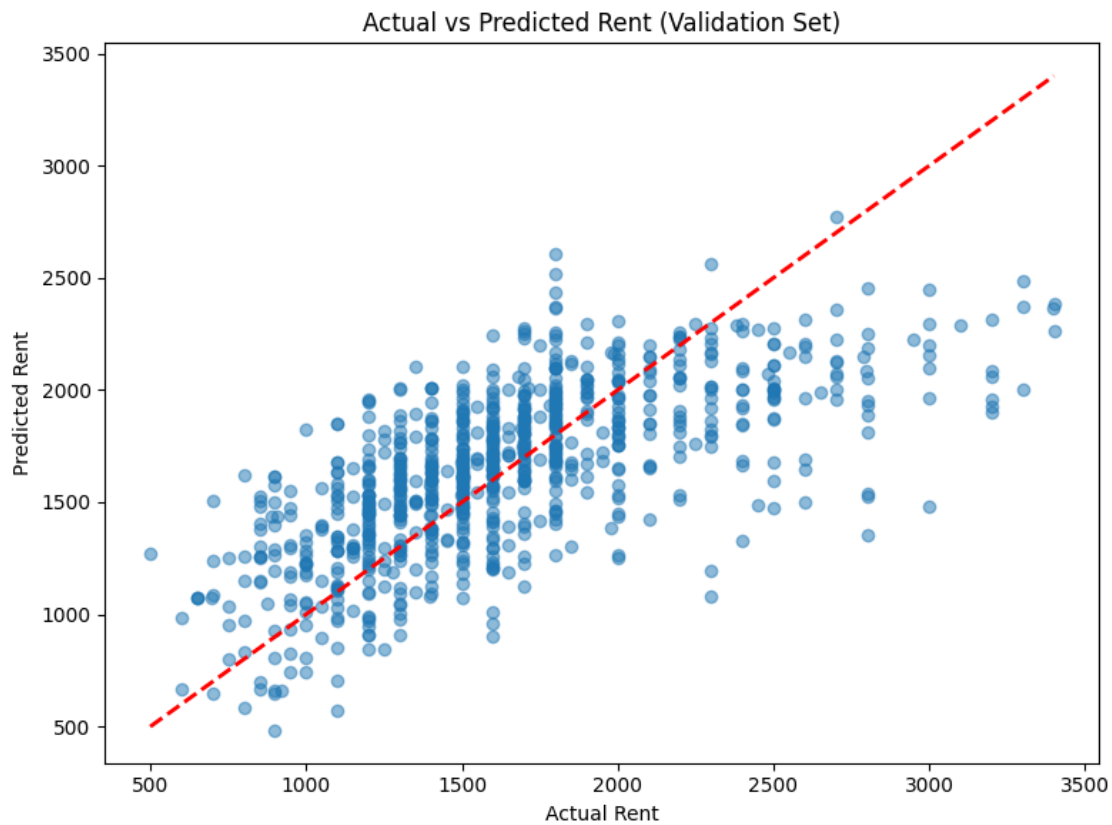
y_pred_val = X_val_clean_b @ theta

mse_val = mean_squared_error(y_val_clean, y_pred_val)
mae_val = mean_absolute_error(y_val_clean, y_pred_val)
r2_val = r2_score(y_val_clean, y_pred_val)

print("Validation Performance")
print(f"Validation MSE: {mse_val:.2f}")
print(f"Validation MAE: {mae_val:.2f}")
print(f"Validation R2: {r2_val:.2f}")
```

Validation Performance
Validation MSE: 148084.08
Validation MAE: 295.19
Validation R^2 : 0.45

```
[34]: plt.figure(figsize=(8, 6))
plt.scatter(y_val_clean, y_pred_val, alpha=0.5)
plt.plot([min(y_val_clean), max(y_val_clean)],
         [min(y_val_clean), max(y_val_clean)],
         color='red', linestyle='--', linewidth=2)
plt.xlabel('Actual Rent')
plt.ylabel('Predicted Rent')
plt.title('Actual vs Predicted Rent (Validation Set)')
plt.tight_layout()
plt.show()
```



```
[ ]:
```

```
[ ]: !apt-get install texlive-xetex
```

Reading package lists... Done

Building dependency tree... Done

Reading state information... Done

The following additional packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
libsyntax2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13
lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
teckit tex-common tex-gyre texlive-base texlive-binaries
texlive-fonts-recommended texlive-latex-base texlive-latex-extra
texlive-latex-recommended texlive-pictures texlive-plain-generic tipa
xfonts-encodings xfonts-utils

Suggested packages:

fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
| postscript-viewer perl-tk xpdf | pdf-viewer xzdec
texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl
texlive-latex-extra-doc texlive-latex-recommended-doc texlive-luatex
texlive-pstricks dot2tex prerex texlive-pictures-doc vprerex
default-jre-headless tipa-doc

The following NEW packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
libsyntax2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13
lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
teckit tex-common tex-gyre texlive-base texlive-binaries
texlive-fonts-recommended texlive-latex-base texlive-latex-extra
texlive-latex-recommended texlive-pictures texlive-plain-generic
texlive-xetex tipa xfonts-encodings xfonts-utils

0 upgraded, 53 newly installed, 0 to remove and 34 not upgraded.

Need to get 182 MB of archives.

After this operation, 571 MB of additional disk space will be used.

Get:1 <http://archive.ubuntu.com/ubuntu/jammy/main amd64 fonts-droid-fallback all 1:6.0.1r16-1.1build1> [1,805 kB]

Get:2 <http://archive.ubuntu.com/ubuntu/jammy/main amd64 fonts-lato all 2.0-2.1> [2,696 kB]

Get:3 <http://archive.ubuntu.com/ubuntu/jammy/main amd64 poppler-data all 0.4.11-1> [2,171 kB]

Get:4 <http://archive.ubuntu.com/ubuntu/jammy/universe amd64 tex-common all 6.17>

[33.7 kB]
Get:5 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-urw-base35 all 20200910-1 [6,367 kB]
Get:6 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libgs9-common all 9.55.0~dfsg1-0ubuntu5.11 [753 kB]
Get:7 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libidn12 amd64 1.38-4ubuntu1 [60.0 kB]
Get:8 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libijs-0.35 amd64 0.35-15build2 [16.5 kB]
Get:9 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libjbig2dec0 amd64 0.19-3build2 [64.7 kB]
Get:10 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libgs9 amd64 9.55.0~dfsg1-0ubuntu5.11 [5,031 kB]
Get:11 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libkpathsea6 amd64 2021.20210626.59705-1ubuntu0.2 [60.4 kB]
Get:12 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libwoff1 amd64 1.0.2-1build4 [45.2 kB]
Get:13 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 dvisvgm amd64 2.13.1-1 [1,221 kB]
Get:14 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-lmodern all 2.004.5-6.1 [4,532 kB]
Get:15 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-noto-mono all 20201225-1build1 [397 kB]
Get:16 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-texgyre all 20180621-3.1 [10.2 MB]
Get:17 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libapache-pom-java all 18-1 [4,720 B]
Get:18 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-parent-java all 43-1 [10.8 kB]
Get:19 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-logging-java all 1.2-2 [60.3 kB]
Get:20 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libptexenc1 amd64 2021.20210626.59705-1ubuntu0.2 [39.1 kB]
Get:21 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rubygems-integration all 1.18 [5,336 B]
Get:22 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby3.0 amd64 3.0.2-7ubuntu2.10 [50.1 kB]
Get:23 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby-rubygems all 3.3.5-2 [228 kB]
Get:24 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby amd64 1:3.0~exp1 [5,100 B]
Get:25 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rake all 13.0.6-2 [61.7 kB]
Get:26 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby-net-telnet all 0.1.1-2 [12.6 kB]
Get:27 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-webrick all 1.7.0-3ubuntu0.1 [52.1 kB]
Get:28 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-xmlrpc all

0.3.2-1ubuntu0.1 [24.9 kB]
 Get:29 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libruby3.0
 amd64 3.0.2-7ubuntu2.10 [5,114 kB]
 Get:30 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libsyntax2
 amd64 2021.20210626.59705-1ubuntu0.2 [55.6 kB]
 Get:31 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libteckit0 amd64
 2.5.11+ds1-1 [421 kB]
 Get:32 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexlua53
 amd64 2021.20210626.59705-1ubuntu0.2 [120 kB]
 Get:33 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexluaajit2
 amd64 2021.20210626.59705-1ubuntu0.2 [267 kB]
 Get:34 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libzip-0-13 amd64
 0.13.72+dfsg.1-1.1 [27.0 kB]
 Get:35 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 xfonts-encodings all
 1:1.0.5-0ubuntu2 [578 kB]
 Get:36 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 xfonts-utils amd64
 1:7.7+6build2 [94.6 kB]
 Get:37 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 lmodern all
 2.004.5-6.1 [9,471 kB]
 Get:38 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 preview-latex-style
 all 12.2-1ubuntu1 [185 kB]
 Get:39 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 t1utils amd64
 1.41-4build2 [61.3 kB]
 Get:40 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 teckit amd64
 2.5.11+ds1-1 [699 kB]
 Get:41 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 tex-gyre all
 20180621-3.1 [6,209 kB]
 Get:42 <http://archive.ubuntu.com/ubuntu> jammy-updates/universe amd64 texlive-
 binaries amd64 2021.20210626.59705-1ubuntu0.2 [9,860 kB]
 Get:43 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 texlive-base all
 2021.20220204-1 [21.0 MB]
 Get:44 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 texlive-fonts-
 recommended all 2021.20220204-1 [4,972 kB]
 Get:45 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 texlive-latex-base
 all 2021.20220204-1 [1,128 kB]
 Get:46 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libfontbox-java all
 1:1.8.16-2 [207 kB]
 Get:47 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libpdfbox-java all
 1:1.8.16-2 [5,199 kB]
 Get:48 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 texlive-latex-
 recommended all 2021.20220204-1 [14.4 MB]
 Get:49 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 texlive-pictures
 all 2021.20220204-1 [8,720 kB]
 Get:50 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 texlive-latex-extra
 all 2021.20220204-1 [13.9 MB]
 Get:51 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 texlive-plain-
 generic all 2021.20220204-1 [27.5 MB]
 Get:52 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 tipa all 2:1.3-21

```

[2,967 kB]
Get:53 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-xetex all
2021.20220204-1 [12.4 MB]
Fetched 182 MB in 8s (21.9 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
Selecting previously unselected package fonts-droid-fallback.
(Reading database ... 126326 files and directories currently installed.)
Preparing to unpack .../00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
...
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Selecting previously unselected package fonts-lato.
Preparing to unpack .../01-fonts-lato_2.0-2.1_all.deb ...
Unpacking fonts-lato (2.0-2.1) ...
Selecting previously unselected package poppler-data.
Preparing to unpack .../02-poppler-data_0.4.11-1_all.deb ...
Unpacking poppler-data (0.4.11-1) ...
Selecting previously unselected package tex-common.
Preparing to unpack .../03-tex-common_6.17_all.deb ...
Unpacking tex-common (6.17) ...
Selecting previously unselected package fonts-urw-base35.
Preparing to unpack .../04-fonts-urw-base35_20200910-1_all.deb ...
Unpacking fonts-urw-base35 (20200910-1) ...
Selecting previously unselected package libgs9-common.
Preparing to unpack .../05-libgs9-common_9.55.0~dfsg1-0ubuntu5.11_all.deb ...
Unpacking libgs9-common (9.55.0~dfsg1-0ubuntu5.11) ...
Selecting previously unselected package libidn12:amd64.
Preparing to unpack .../06-libidn12_1.38-4ubuntu1_amd64.deb ...
Unpacking libidn12:amd64 (1.38-4ubuntu1) ...
Selecting previously unselected package libijs-0.35:amd64.
Preparing to unpack .../07-libijs-0.35_0.35-15build2_amd64.deb ...
Unpacking libijs-0.35:amd64 (0.35-15build2) ...
Selecting previously unselected package libjbig2dec0:amd64.
Preparing to unpack .../08-libjbig2dec0_0.19-3build2_amd64.deb ...
Unpacking libjbig2dec0:amd64 (0.19-3build2) ...
Selecting previously unselected package libgs9:amd64.
Preparing to unpack .../09-libgs9_9.55.0~dfsg1-0ubuntu5.11_amd64.deb ...
Unpacking libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.11) ...
Selecting previously unselected package libkpathsea6:amd64.
Preparing to unpack .../10-libkpathsea6_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack .../11-libwoff1_1.0.2-1build4_amd64.deb ...
Unpacking libwoff1:amd64 (1.0.2-1build4) ...
Selecting previously unselected package dvisvgm.
Preparing to unpack .../12-dvisvgm_2.13.1-1_amd64.deb ...
Unpacking dvisvgm (2.13.1-1) ...

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```

Selecting previously unselected package fonts-lmodern.
Preparing to unpack .../13-fonts-lmodern_2.004.5-6.1_all.deb ...
Unpacking fonts-lmodern (2.004.5-6.1) ...
Selecting previously unselected package fonts-noto-mono.
Preparing to unpack .../14-fonts-noto-mono_20201225-1build1_all.deb ...
Unpacking fonts-noto-mono (20201225-1build1) ...
Selecting previously unselected package fonts-texgyre.
Preparing to unpack .../15-fonts-texgyre_20180621-3.1_all.deb ...
Unpacking fonts-texgyre (20180621-3.1) ...
Selecting previously unselected package libapache-pom-java.
Preparing to unpack .../16-libapache-pom-java_18-1_all.deb ...
Unpacking libapache-pom-java (18-1) ...
Selecting previously unselected package libcommons-parent-java.
Preparing to unpack .../17-libcommons-parent-java_43-1_all.deb ...
Unpacking libcommons-parent-java (43-1) ...
Selecting previously unselected package libcommons-logging-java.
Preparing to unpack .../18-libcommons-logging-java_1.2-2_all.deb ...
Unpacking libcommons-logging-java (1.2-2) ...
Selecting previously unselected package libptexenc1:amd64.
Preparing to unpack .../19-libptexenc1_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package rubygems-integration.
Preparing to unpack .../20-rubygems-integration_1.18_all.deb ...
Unpacking rubygems-integration (1.18) ...
Selecting previously unselected package ruby3.0.
Preparing to unpack .../21-ruby3.0_3.0.2-7ubuntu2.10_amd64.deb ...
Unpacking ruby3.0 (3.0.2-7ubuntu2.10) ...
Selecting previously unselected package ruby-rubygems.
Preparing to unpack .../22-ruby-rubygems_3.3.5-2_all.deb ...
Unpacking ruby-rubygems (3.3.5-2) ...
Selecting previously unselected package ruby.
Preparing to unpack .../23-ruby_1%3a3.0~exp1_amd64.deb ...
Unpacking ruby (1:3.0~exp1) ...
Selecting previously unselected package rake.
Preparing to unpack .../24-rake_13.0.6-2_all.deb ...
Unpacking rake (13.0.6-2) ...
Selecting previously unselected package ruby-net-telnet.
Preparing to unpack .../25-ruby-net-telnet_0.1.1-2_all.deb ...
Unpacking ruby-net-telnet (0.1.1-2) ...
Selecting previously unselected package ruby-webrick.
Preparing to unpack .../26-ruby-webrick_1.7.0-3ubuntu0.1_all.deb ...
Unpacking ruby-webrick (1.7.0-3ubuntu0.1) ...
Selecting previously unselected package ruby-xmlrpc.
Preparing to unpack .../27-ruby-xmlrpc_0.3.2-1ubuntu0.1_all.deb ...
Unpacking ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Selecting previously unselected package libruby3.0:amd64.
Preparing to unpack .../28-libruby3.0_3.0.2-7ubuntu2.10_amd64.deb ...

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Unpacking libruby3.0:amd64 (3.0.2-7ubuntu2.10) ...
Selecting previously unselected package libsyntax2:amd64.
Preparing to unpack .../29-libsyntax2_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libsyntax2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libteckit0:amd64.
Preparing to unpack .../30-libteckit0_2.5.11+ds1-1_amd64.deb ...
Unpacking libteckit0:amd64 (2.5.11+ds1-1) ...
Selecting previously unselected package libtexlua53:amd64.
Preparing to unpack .../31-libtexlua53_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libtexluajit2:amd64.
Preparing to unpack
.../32-libtexluajit2_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libzzip-0-13:amd64.
Preparing to unpack .../33-libzzip-0-13_0.13.72+dfsg.1-1.1_amd64.deb ...
Unpacking libzzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Selecting previously unselected package xfonts-encodings.
Preparing to unpack .../34-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb ...
Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) ...
Selecting previously unselected package xfonts-utils.
Preparing to unpack .../35-xfonts-utils_1%3a7.7+6build2_amd64.deb ...
Unpacking xfonts-utils (1:7.7+6build2) ...
Selecting previously unselected package lmodern.
Preparing to unpack .../36-lmodern_2.004.5-6.1_all.deb ...
Unpacking lmodern (2.004.5-6.1) ...
Selecting previously unselected package preview-latex-style.
Preparing to unpack .../37-preview-latex-style_12.2-1ubuntu1_all.deb ...
Unpacking preview-latex-style (12.2-1ubuntu1) ...
Selecting previously unselected package t1utils.
Preparing to unpack .../38-t1utils_1.41-4build2_amd64.deb ...
Unpacking t1utils (1.41-4build2) ...
Selecting previously unselected package teckit.
Preparing to unpack .../39-teckit_2.5.11+ds1-1_amd64.deb ...
Unpacking teckit (2.5.11+ds1-1) ...
Selecting previously unselected package tex-gyre.
Preparing to unpack .../40-tex-gyre_20180621-3.1_all.deb ...
Unpacking tex-gyre (20180621-3.1) ...
Selecting previously unselected package texlive-binaries.
Preparing to unpack .../41-texlive-
binaries_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package texlive-base.
Preparing to unpack .../42-texlive-base_2021.20220204-1_all.deb ...
Unpacking texlive-base (2021.20220204-1) ...
Selecting previously unselected package texlive-fonts-recommended.

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Preparing to unpack .../43-texlive-fonts-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-fonts-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-base.
Preparing to unpack .../44-texlive-latex-base_2021.20220204-1_all.deb ...
Unpacking texlive-latex-base (2021.20220204-1) ...
Selecting previously unselected package libfontbox-java.
Preparing to unpack .../45-libfontbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libfontbox-java (1:1.8.16-2) ...
Selecting previously unselected package libpdfbox-java.
Preparing to unpack .../46-libpdfbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libpdfbox-java (1:1.8.16-2) ...
Selecting previously unselected package texlive-latex-recommended.
Preparing to unpack .../47-texlive-latex-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-latex-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive-pictures.
Preparing to unpack .../48-texlive-pictures_2021.20220204-1_all.deb ...
Unpacking texlive-pictures (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-extra.
Preparing to unpack .../49-texlive-latex-extra_2021.20220204-1_all.deb ...
Unpacking texlive-latex-extra (2021.20220204-1) ...
Selecting previously unselected package texlive-plain-generic.
Preparing to unpack .../50-texlive-plain-generic_2021.20220204-1_all.deb ...
Unpacking texlive-plain-generic (2021.20220204-1) ...
Selecting previously unselected package tipa.
Preparing to unpack .../51-tipa_2%3a1.3-21_all.deb ...
Unpacking tipa (2:1.3-21) ...
Selecting previously unselected package texlive-xetex.
Preparing to unpack .../52-texlive-xetex_2021.20220204-1_all.deb ...
Unpacking texlive-xetex (2021.20220204-1) ...
Setting up fonts-lato (2.0-2.1) ...
Setting up fonts-noto-mono (20201225-1build1) ...
Setting up libwoff1:amd64 (1.0.2-1build4) ...
Setting up libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libijs-0.35:amd64 (0.35-15build2) ...
Setting up libtexluaajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libfontbox-java (1:1.8.16-2) ...
Setting up rubygems-integration (1.18) ...
Setting up libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Setting up fonts-urw-base35 (20200910-1) ...
Setting up poppler-data (0.4.11-1) ...
Setting up tex-common (6.17) ...
update-language: texlive-base not installed and configured, doing nothing!
Setting up libbig2dec0:amd64 (0.19-3build2) ...
Setting up libteckit0:amd64 (2.5.11+ds1-1) ...
Setting up libapache-pom-java (18-1) ...
Setting up ruby-net-telnet (0.1.1-2) ...
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) ...
Setting up t1utils (1.41-4build2) ...

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Setting up libidn12:amd64 (1.38-4ubuntu1) ...
Setting up fonts-texgyre (20180621-3.1) ...
Setting up libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up ruby-webrick (1.7.0-3ubuntu0.1) ...
Setting up fonts-lmodern (2.004.5-6.1) ...
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Setting up ruby-xmldr (0.3.2-1ubuntu0.1) ...
Setting up libsynchronet2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libgs9-common (9.55.0~dfsg1-0ubuntu5.11) ...
Setting up teckit (2.5.11+ds1-1) ...
Setting up libpdfbox-java (1:1.8.16-2) ...
Setting up libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.11) ...
Setting up preview-latex-style (12.2-1ubuntu1) ...
Setting up libcommons-parent-java (43-1) ...
Setting up dvisvgm (2.13.1-1) ...
Setting up libcommons-logging-java (1.2-2) ...
Setting up xfonts-utils (1:7.7+6build2) ...
Setting up libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin
(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.004.5-6.1) ...
Setting up texlive-base (2021.20220204-1) ...
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST...
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN...
mktexlsr: Updating /var/lib/texmf/ls-R...
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps
tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-
ini-files/pdftexconfig.tex
Setting up tex-gyre (20180621-3.1) ...
Setting up texlive-plain-generic (2021.20220204-1) ...
Setting up texlive-latex-base (2021.20220204-1) ...
Setting up texlive-latex-recommended (2021.20220204-1) ...
Setting up texlive-pictures (2021.20220204-1) ...
Setting up texlive-fonts-recommended (2021.20220204-1) ...
Setting up tipa (2:1.3-21) ...
Setting up texlive-latex-extra (2021.20220204-1) ...

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Setting up texlive-xetex (2021.20220204-1) ...
Setting up rake (13.0.6-2) ...
Setting up libruby3.0:amd64 (3.0.2-7ubuntu2.10) ...
Setting up ruby3.0 (3.0.2-7ubuntu2.10) ...
Setting up ruby (1:3.0~exp1) ...
Setting up ruby-rubygems (3.3.5-2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for mailcap (3.70+nmu1ubuntu1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libumf.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

Processing triggers for tex-common (6.17) ...
Running updmap-sys. This may take some time... done.
Running mktexlsr /var/lib/texmf ... done.
Building format(s) --all.
    This may take some time...

```

```

[40]: !jupyter nbconvert --to pdf "/content/drive/MyDrive/linear/Linear_regression.
      ↪ipynb"

```

```

[NbConvertApp] Converting notebook
/content/drive/MyDrive/linear/Linear_regression.ipynb to pdf
/usr/local/share/jupyter/nbconvert/templates/latex/display_priority.j2:32:
UserWarning: Your element with mimetype(s) dict_keys(['text/html']) is not able
to be represented.
    ((*- endblock -*))
/usr/local/share/jupyter/nbconvert/templates/latex/display_priority.j2:32:
UserWarning: Your element with mimetype(s) dict_keys(['text/html']) is not able
to be represented.
    ((*- endblock -*))
[NbConvertApp] Support files will be in Linear_regression_files/
[NbConvertApp] Making directory ./Linear_regression_files
[NbConvertApp] Writing 141278 bytes to notebook.tex
[NbConvertApp] Building PDF
Traceback (most recent call last):
  File "/usr/local/bin/jupyter-nbconvert", line 10, in <module>
    sys.exit(main())
    ~~~~~
  File "/usr/local/lib/python3.11/dist-packages/jupyter_core/application.py",
line 283, in launch_instance
    super().launch_instance(argv=argv, **kwargs)
  File "/usr/local/lib/python3.11/dist-
packages/traitlets/config/application.py", line 992, in launch_instance
    app.start()
  File "/usr/local/lib/python3.11/dist-packages/nbconvert/nbconvertapp.py", line
420, in start
    self.convert_notebooks()
  File "/usr/local/lib/python3.11/dist-packages/nbconvert/nbconvertapp.py", line
597, in convert_notebooks
    self.convert_single_notebook(notebook_filename)
  File "/usr/local/lib/python3.11/dist-packages/nbconvert/nbconvertapp.py", line
563, in convert_single_notebook
    output, resources = self.export_single_notebook(
    ~~~~~
  File "/usr/local/lib/python3.11/dist-packages/nbconvert/nbconvertapp.py", line
487, in export_single_notebook
    output, resources = self.exporter.from_filename(
    ~~~~~
  File "/usr/local/lib/python3.11/dist-
packages/nbconvert/exporters/templateexporter.py", line 390, in from_filename
    return super().from_filename(filename, resources, **kw) #
type:ignore[return-value]
    ~~~~~
  File "/usr/local/lib/python3.11/dist-
packages/nbconvert/exporters/exporter.py", line 201, in from_filename
    return self.from_file(f, resources=resources, **kw)
    ~~~~~
  File "/usr/local/lib/python3.11/dist-

```



```

packages/nbconvert/exporters/templateexporter.py", line 396, in from_file
    return super().from_file(file_stream, resources, **kw) #
type:ignore[return-value]
~~~~~

File "/usr/local/lib/python3.11/dist-
packages/nbconvert/exporters/exporter.py", line 220, in from_file
    return self.from_notebook_node(
~~~~~

File "/usr/local/lib/python3.11/dist-packages/nbconvert/exporters/pdf.py",
line 197, in from_notebook_node
    self.run_latex(tex_file)
File "/usr/local/lib/python3.11/dist-packages/nbconvert/exporters/pdf.py",
line 166, in run_latex
    return self.run_command(
~~~~~

File "/usr/local/lib/python3.11/dist-packages/nbconvert/exporters/pdf.py",
line 120, in run_command
    raise OSError(msg)
OSError: xelatex not found on PATH, if you have not installed xelatex you may
need to do so. Find further instructions at
https://nbconvert.readthedocs.io/en/latest/install.html#installing-tex.

```